



MASTER IN ASTROPHYSICS
AND SPACE SCIENCE



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Radiation Physics for SMBH Accretion Disks in PLUTO

CAMK PAN SUMMER STUDENT PROGRAM

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Completed Work

- Modeled Schwarzschild BH in Viscous HD, MHD and Resistive MHD (Axial & Equatorial Symmetry)
 - Replaced Newtonian with Paczyński–Wiita Potential to Reproduce ISCO

$$\phi_N(r) \rightarrow \phi_{PW}(r) = -\frac{GM}{r - r_S}$$

- Rescaled Code Units to Gravitational Radii and Timescales as Convention

$$r_G = GM/c^2 \qquad \tau_G = GM/c^3$$

- Inner Boundary: Outflow with Backflow Preventing Diode Condition
- Outer Boundary: Extrapolation with Van Leer Limited Slope, Coronal Backflow Guard and Injected Kluźniak-Kita Analytic Disk Solution

Completed Work

- Implemented Physically Motivated Replacement for Passive Tracer of Disk and Corona
 - Advected Tracer Field as Switch for Viscosity, Resistivity, Opacity

$$\partial_t q + \mathbf{v} \cdot \nabla q = 0$$

- Native Implementation Fails if Radiation Physics Enabled
- Fix?

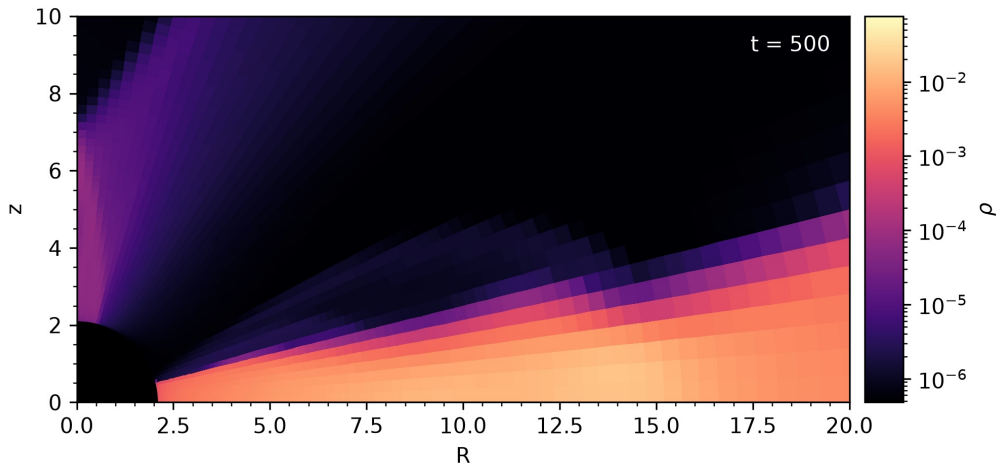
Completed Work

- Implemented Physically Motivated Replacement for Passive Tracer of Disk and Corona
 - Reconstruct Smooth Classifier $f \in [0, 1]$ from Density and Rotation
 - Calculate Adaptive Disk and Corona Weighted Mean Density Profiles
 - Compute Time Smoothed Radial Reference via Exponential Moving Average

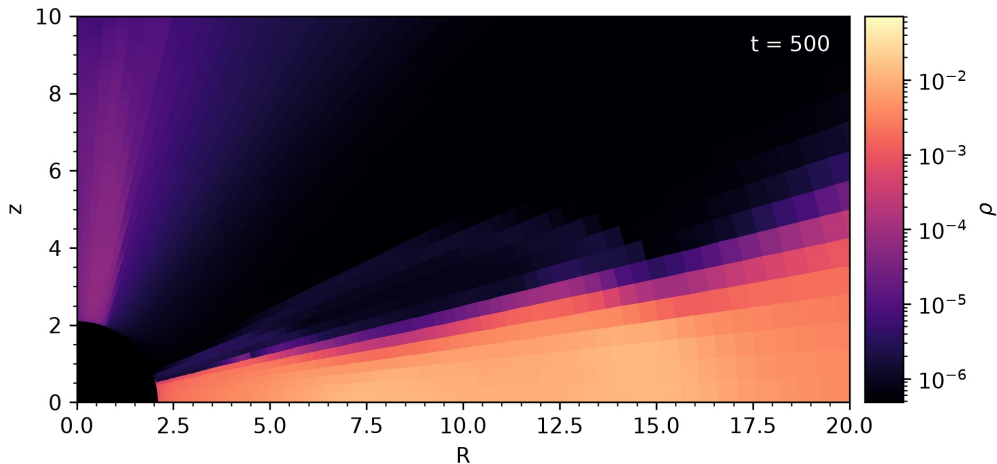
$$\rho_n = \alpha \tilde{\rho}_n + (1 - \alpha) \rho_{n-1}$$

- Obtain Discriminator f_ρ by Feeding Cell Density into Sigmoid
 - Similarly Find f_w from Comparing Local Azimuthal to Keplerian Velocity
 - Multiply like $f = f_\rho \cdot f_w$ for Disk and $1 - f$ for Corona Weights
 - Account for Initial Conditions (Truncated Disk & Hard Binary)
- Parallelize with MPI Reduction Across Entire Domain and Feeding Back to All Ranks

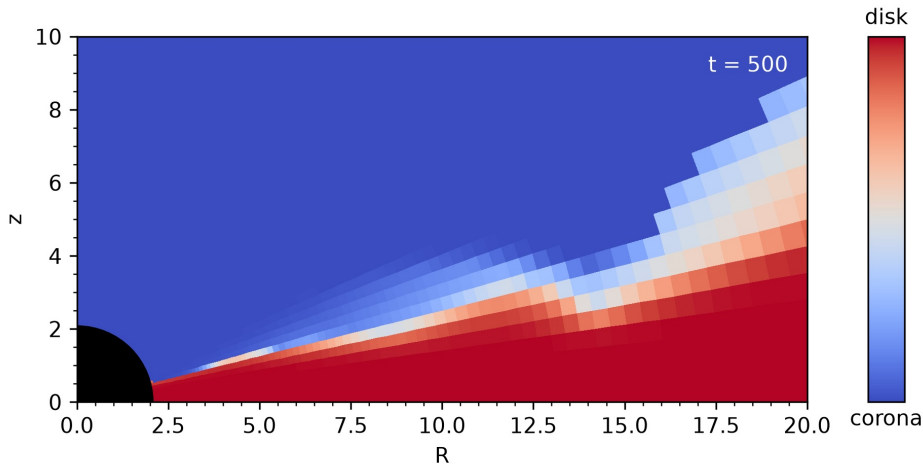
Tracer Physics – Density



Recovered Physics – Density

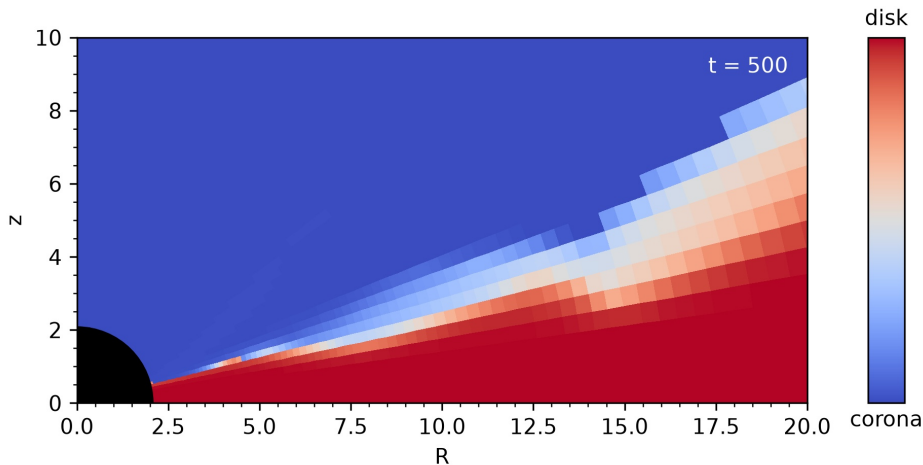


Tracer Physics – Tracer

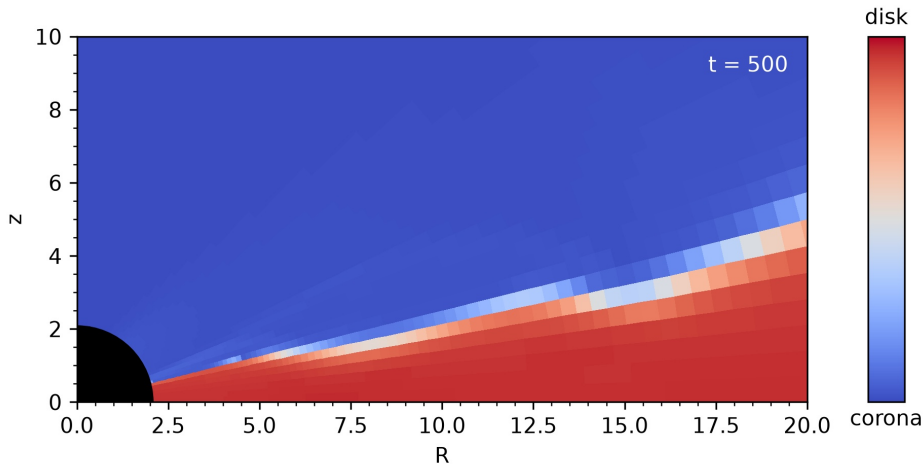


Graphical Comparison

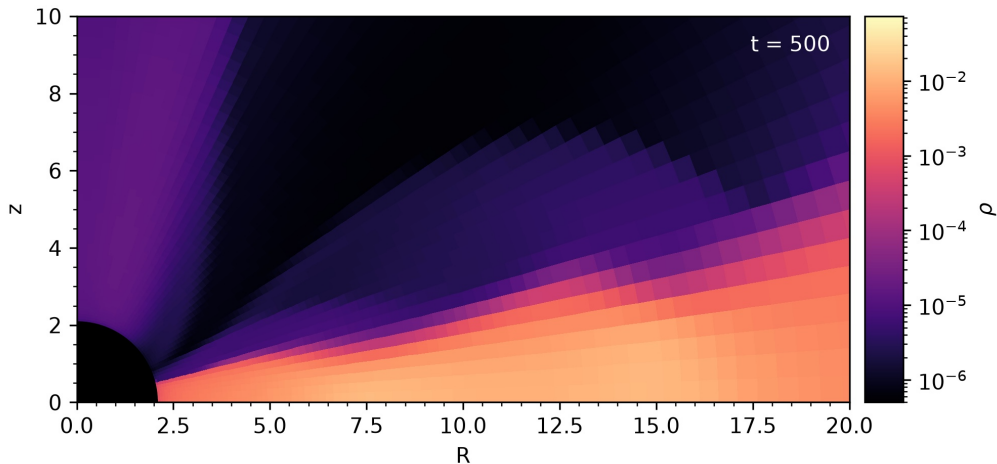
Recovered Physics – Tracer



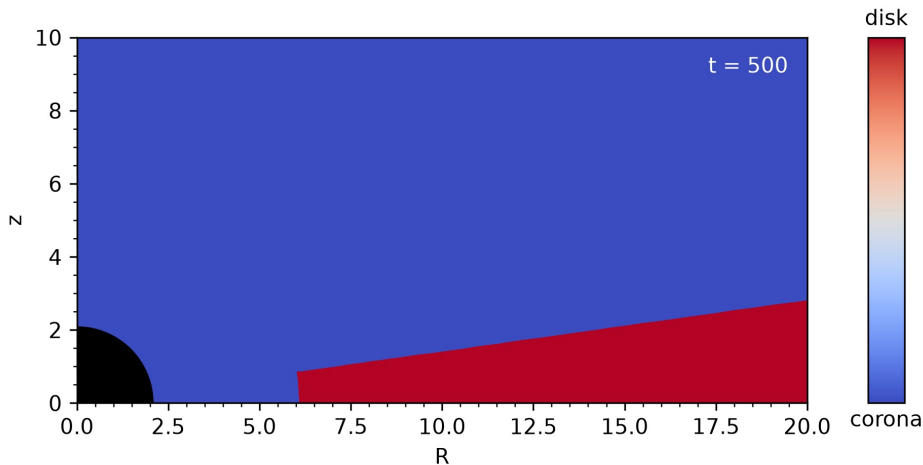
Recovered Physics – Recovered



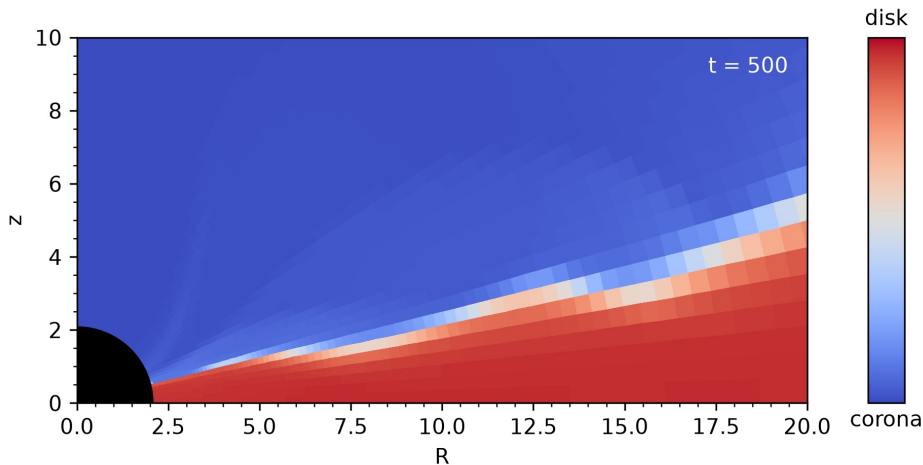
Radiation Enabled – Density



Radiation Enabled – Tracer



Radiation Enabled – Recovered



Next Steps

- Verify Correct Physical Units and Blackbody Radiation Field

$$\kappa_{\text{scat}} = f \cdot \rho \cdot \kappa_{\text{es}} \qquad \kappa_{\text{abs}} = f \cdot \rho \cdot (\kappa_{\text{bf}} + \kappa_{\text{ff}})$$

- Implement SMBH Scaling and Quantify Radiative Transfer Effects
- Enable Ideal or Resistive MHD Solver and Test Magnetic Field Interactions
- Extend 2.5D to 3D Simulations and Apply Radiation Setup in Different Scenarios

Thank You!

- A. Mignone et al., *PLUTO: A Numerical Code for Computational Astrophysics* (ApJ) 2007
- M. Čemeljić, “Atlas” of Numerical Solutions for Star-Disk Magnetospheric Interaction (A&A) 2019
- M. Čemeljić, W. Kluźniak, S. Kundu, *Magnetic Precession as a Model for Quasiperiodic Oscillations in Pulsating Ultra Luminous X-Ray and Flat-Top Noise Sources* (A&A) 2025

Resources on GitHub: [here](#)